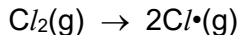


8 Assuming the pressure of each system remains at 1 atm, which processes show an increase in entropy?

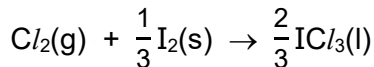
1 1 mol of $\text{Cl}_2(\text{g})$ at 298 K is heated to 373 K.

2 1 mol of $\text{Cl}_2(\text{g})$ at 298 K is added to 1 mol of $\text{N}_2(\text{g})$ at 298 K.

3 1 mol of $\text{Cl}_2(\text{g})$ at 1000 K is photolysed according to the following equation.



4 1 mol of $\text{Cl}_2(\text{g})$ at 298 K is reacted with iodine according to the following equation.



A 1, 2 and 3 only

B 1, 2 and 4 only

C 1, 3 and 4 only

D 2, 3 and 4 only